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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Experiment – 7: Database as a Service(DBaaS)

Aim: To create and Connect to a MySQL Database with Amazon RDS. (Free Tier Cloud Platform).

Objective: After performing the experiment, the students will be able to learn –
- Create an environment to run MySQL database
- Connect to the database
- How To Access RDS Database
- Delete the database instance

Lab objective mapped : ITDO8024.3 : To learn and examine cloud service model and deploy database, storage services over different cloud platforms.

Prerequisite:
An AWS account,
MySQL Workbench application.

Requirements: Cloud Login**,** Desktop, Browser, Internet, GitHub etc.

Pre-Experiment Theory: (describe following topics in soft copy form)

1. **Database services-**

Sr No	Database type		AWS service
1	Relational Database	Relational databases store data with predefined schemas and relationships between them. These databases are designed to support ACID transactions, and maintain referential integrity and strong data consistency.	1. Amazon Aurora 2. Amazon RDS 3. Amazon Redshift
2	In-memory Database	In-memory databases are used for applications that require real-time access to data. By storing data directly in memory, these databases deliver microsecond latency to applications for whom millisecond latency is not enough.	1. Amazon ElastiCache 2. Amazon MemoryDB for Redis

3	Document Database	A document database is designed to store semistructured data as JSON-like documents. These databases help developers build and update applications quickly.	Amazon DocumentDB (with MongoDB compatibility)
4	Key-value Database	Key-value databases are optimized for common access patterns, typically to store and retrieve large volumes of data. These databases deliver quick response times, even in extreme volumes of concurrent requests.	Amazon DynamoDB

2. **Amazon RDS**
   Amazon RDS is an easy to manage relational database service optimized for total cost of ownership. It is simple to set up, operate, and scale with demand. Amazon RDS automates the undifferentiated database management tasks, such as provisioning, configuring, backups, and patching. Amazon RDS enables customers to create a new database in minutes, and offers flexibility to customize databases to meet their needs across 8 engines and 2 deployment options. Customers can optimize performance with features, like Multi-AZ with two readable standbys, Optimized Writes and Reads, and AWS Graviton3-based instances, and choose from multiple pricing options to effectively manage costs.

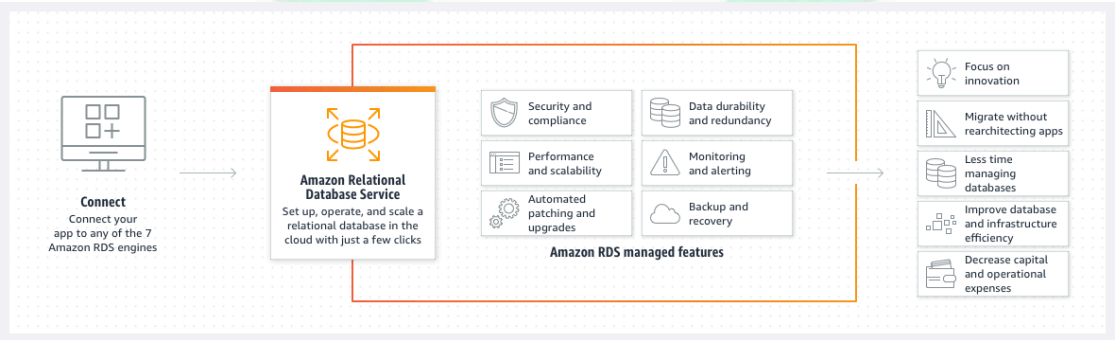


Fig. Amazon Relational Database Service (Amazon RDS) working

Procedure

Step1: Create MY SQL DB instance

1. select AWS RDS service
2. change to N.vergina region
3. select option ‘create database’
4. select option ‘standard create’ and not ‘easy create’
5. select engine option as MYSQL
   edition- MYSQL community
   engine version -MYSQL (some version number)
   Template: free tier
   availability and durability - all disable
6. DB instance configuration steps:
   settings:

DB instance identifier - Name DB
Master uname:- admin
Master pwd: (give some pwd)
confirm pwd: (retype)

7. Instance configuration:
   DB instance class: Burstable classes
   select db.t2.micro
8. Instance specifications:
   storage type: General purpose (SSD)
   allocated storage- 20 GB
   enable storage autoscaling - disable
9. connectivity
   compute resource: -Don't connect to an EC2 compute resource
   VPC -default
   subnet - default
10. Public accessibility: yes RDS assignes a public IP address.
    (This setting will enable direct connection to the database from your device)
11. VPC security group:
    create new VPC security group (this will allow connection from the IP address of the device to created database)
    Name new security group
12. availability zone: No preference
13. RDS proxy - unchecked
14. Additional configuration:
    Port -3306 (default)
15. Database Authentication: select pwd authentication
16. Monitoring:
    Enhanced monitoring unchecked [paid service- allow to check metrics in real time for the os that DB instance runs on]
17. Additional Configuration:
    Database options
    name- dbname(compulsary) AWS RDS does not create it automatically if not specified.
    DB parameter group - default
    option group - default
    Encryption -not for free
    Back up - retention period 1 day
    Back up window- no preference
    copy tag to snapshot - checked
    log exports - all unchecked
    IAM role -default
    Maintainance:
    enable auto mirror - checked
    maintenance window - no preference
    Deletion protection: enable -unchecked
18. select Create Database
    DB is being created.

Status changing to Creating to backing-up to Available (indicating DB is running)

Step2: Download SQL workbench client

Download MYSQL Workbench client and install . (No sign-up required)

Step3: Connect to MySQL DB using SQL workbench client

Launch MySQLWorkbench application.

Select Database - connect to database

Enter the following:

Hostname : endpoint info of RDS

Port : 3306

Username and pwd: Use the ones set earlier while creating AWS RDS DB.

Pwd stored in vault - ok

Run the following SQL quires on SQL workbench client

To execute the SQL statement highlight the statement from the coding window and execute. Observe the action output.

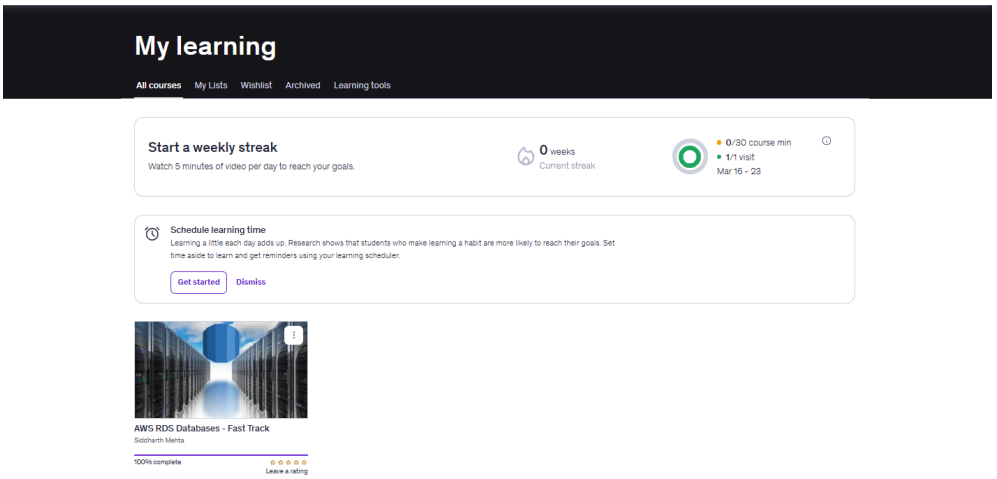
```
CREATE  DATABASE ex;
USE ex;
CREATE TABLE st(
  table_id INT PRIMARY KEY,
  stud_id  INT,
  stud_name CHAR(10));
desc st;
INSERT INTO st( table_id, stud_id, Stud_name) values (01, 100, xyz)
SELECT * FROM st;
DELETE FROM st WHERE table=01;
DROP TABLE st;
```

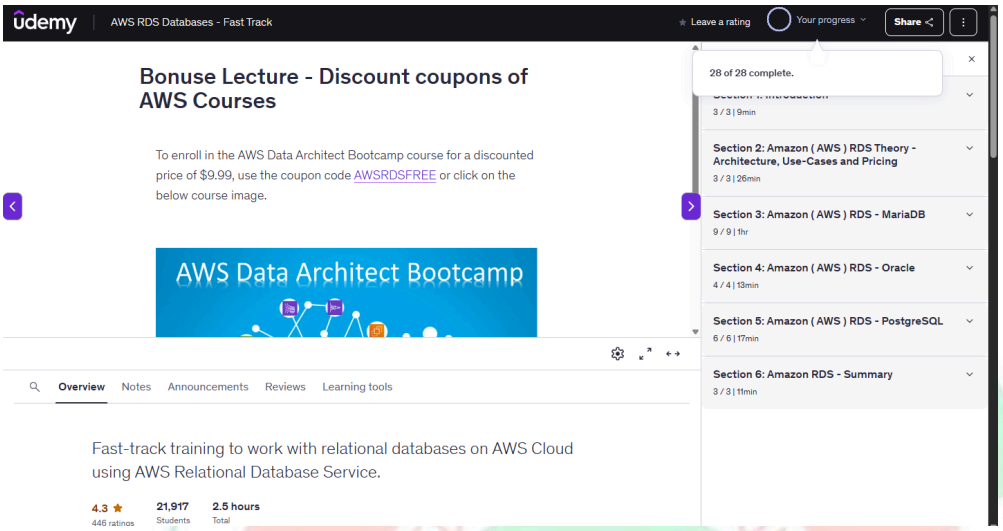
Step 4: Clean up DB instance

Delete the DB instance and security group created inside AWS RDS.

Post-Experiments Exercise

AWS RDS Databases - Fast Track (free course on Udemy):
https://www.udemy.com/course/aws-rds-databases-tutorial-training-free/
complete the course and attach completion proof (like enrollemnet proof, prepare one google_quiz_10mrks_attach link here, proof of notes taken etc)





Extended Theory:

1. Why are databases essential?(to be written in hand)
2. How many DB instances can I run with Amazon RDS?(to be written in hand)
3. Types of AWS Databases(Complete given table)(soft copy)

Database Providers	AWS	AZURE	GOOGLE
NAMES	1. Amazon Aurora 2. Amazon Redshift 3. Amazon RDS	1. Azure SQL Database 2. Azure Database for MySQL 3. Azure Cosmos DB	1. Cloud SQL 2. Cloud Spanner 3. BigQuery

Results/Calculations/Observations:

Fill the following observation tables-(to be written in hand)

Amazon Aurora Vs RDS

Sr. No	**Parameter**	**Amazon Aurora**	**Amazon RDS**
1.	Performance		
2.	Compatibility with Database Engines		
3.	Limitations		

Post Experimental Exercise-(soft copy)

Questions:

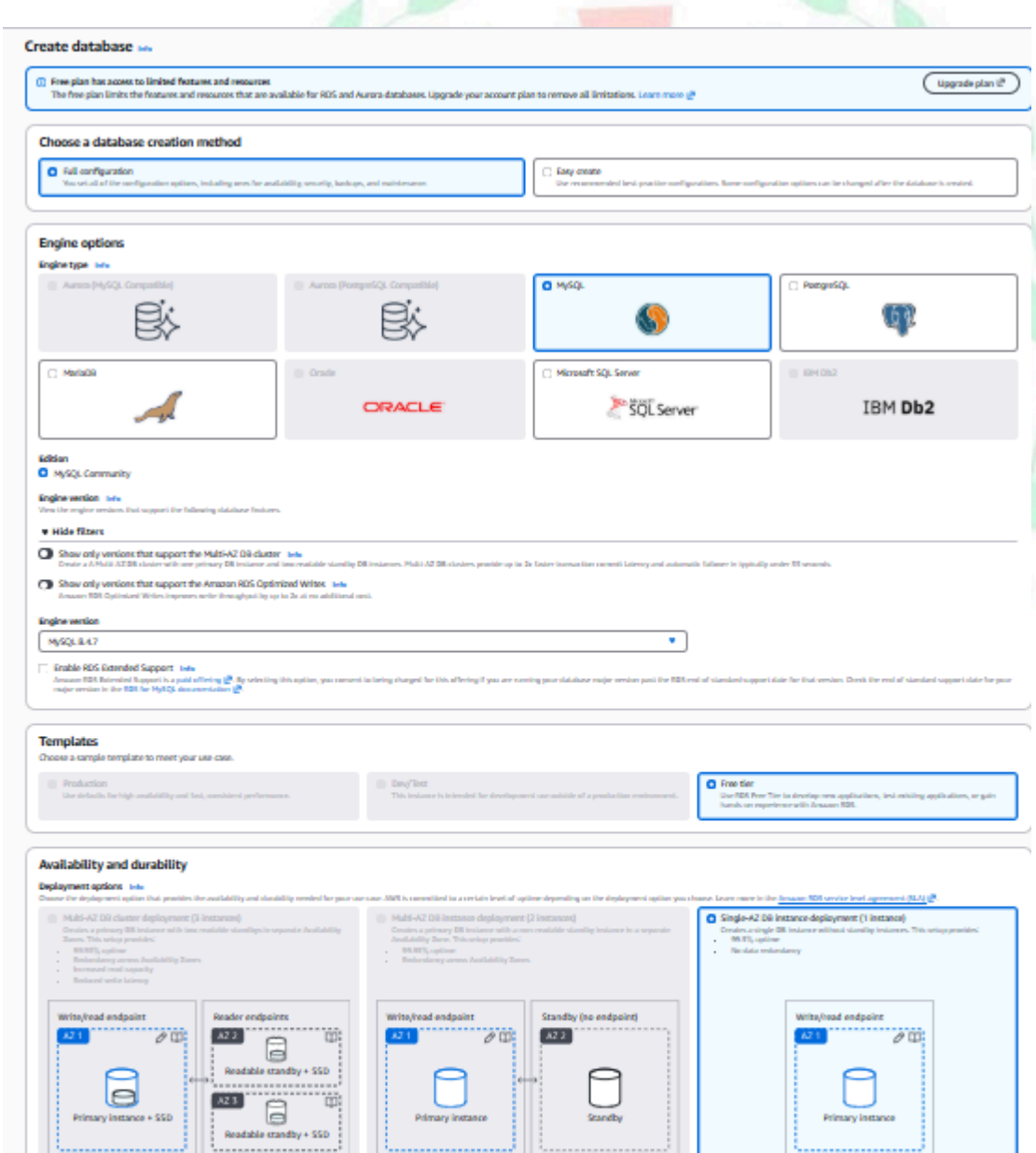
1. What is a database instance (DB instance)?
   A **database instance (DB instance)** in Amazon RDS is an isolated database environment in the cloud where a database engine such as MySQL, PostgreSQL, or MariaDB runs. It contains the compute resources, memory, storage, and network settings required to operate the database. A DB instance allows users to create, store, manage, and access databases without manually handling tasks like

hardware provisioning, patching, backups, or scaling, since these operations are managed automatically by Amazon RDS.

2. How will I be charged and billed for my use of Amazon RDS?
   Amazon RDS follows a **pay-as-you-go pricing model**, where users are charged based on the resources they consume. Charges mainly depend on factors such as the type and size of the DB instance, the amount of storage allocated, backup storage, and additional features like multi-AZ deployment or monitoring. Data transfer and snapshot storage may also incur costs. If the usage stays within the AWS **Free Tier limits**, users are not charged; otherwise, billing is calculated according to the selected instance configuration and usage duration.

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:
[1]
[Online]https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Migrating.RDSMySQL.Import.html#AuroraMySQL.Migrating.RDSMySQL.Space
[2] [Online]https://docs.aws.amazon.com/Am azonRDS/latest/UserGuide/Welcome.html



admin

Credentials management

Managed in AWS Secrets Manager - most secure

Self managed

Auto generate password

Master password | Info

Password strength Very weak

Confirm master password | Info

Additional credentials settings

Database authentication options Info

Password authentication

Password and IAM database authentication

Password and Kerberos authentication

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class | Info

Hide filters

Show instance classes that support Amazon RDS Optimized Writes Info

Include previous generation classes

Standard classes (includes m classes)

Memory optimized classes (includes r and x classes)

Burstable classes (includes t classes)

Instance type

db.t4g.micro

Storage

Storage type Info

General Purpose SSD (gp2)

Allocated storage Info

20 GiB

Additional storage configuration

Storage autoscaling | Info

Enable storage autoscaling

Connectivity Info

Compute resource

Don't connect to an EC2 compute resource

Connect to an EC2 compute resource

Virtual private cloud (VPC) Info

Default VPC (vpc-0222fddf13adfddf729)

After a database is created, you can't change its VPC.

DB subnet group Info

default

Public access Info

Yes

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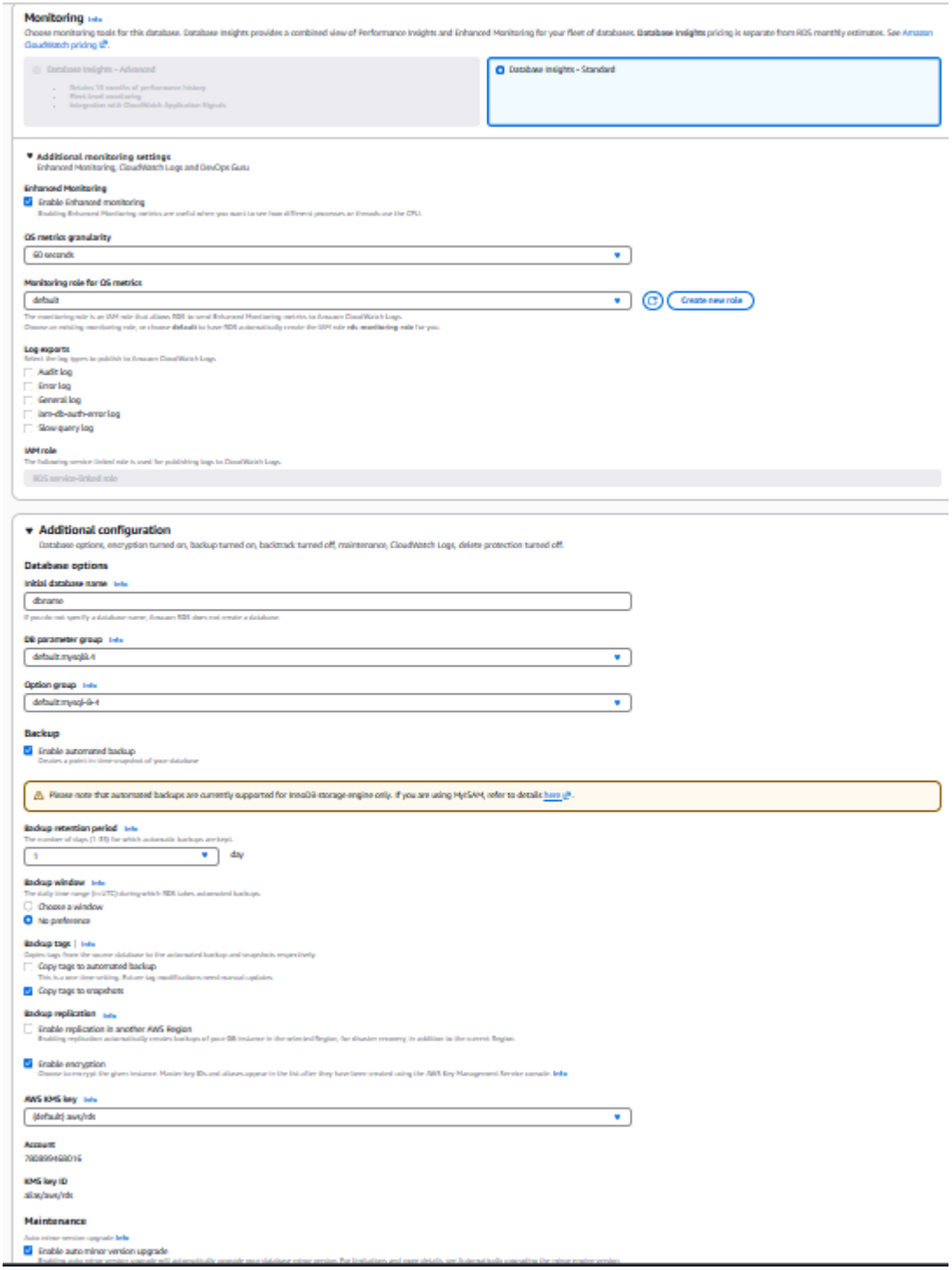
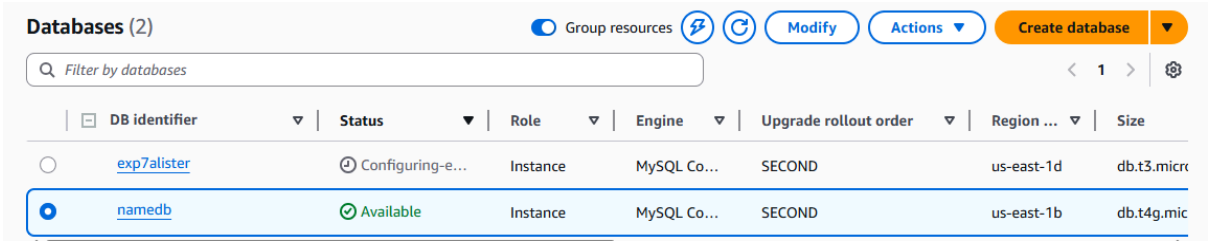
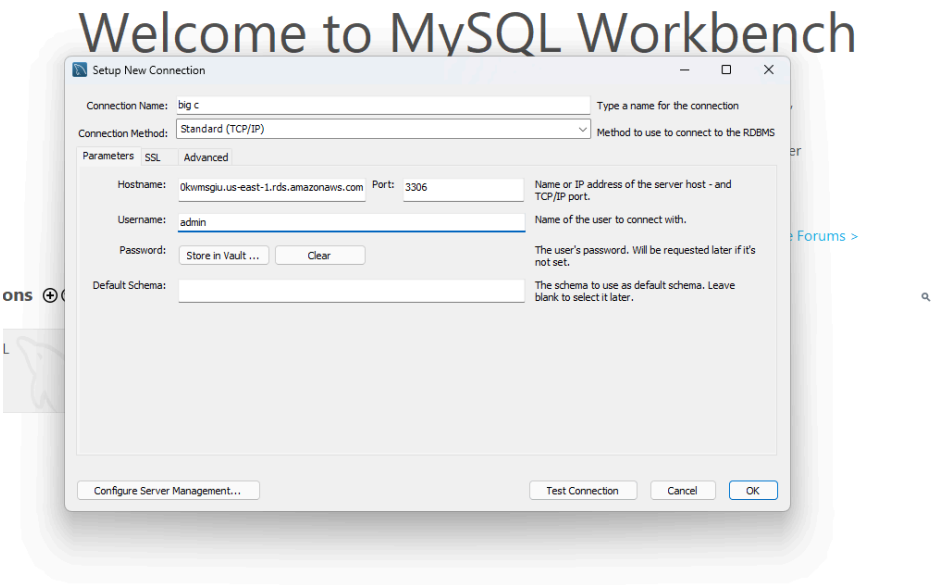
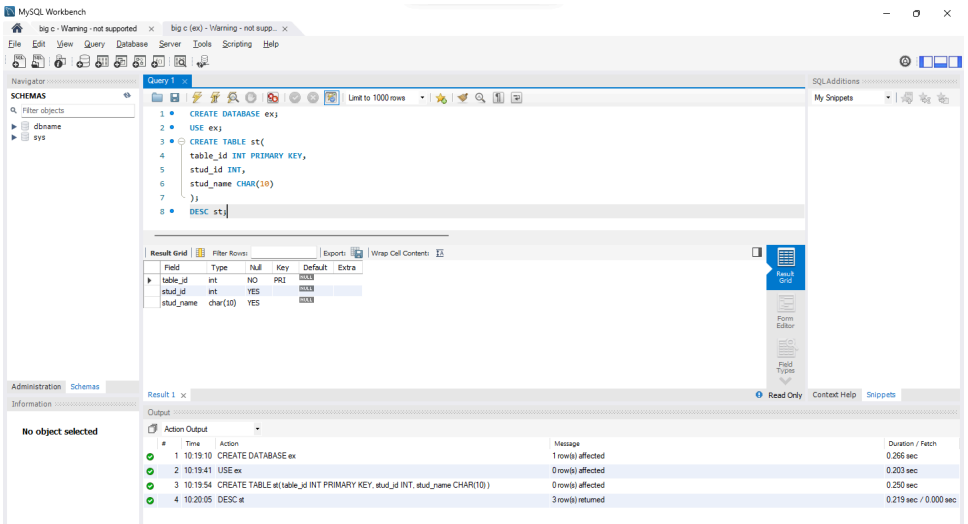





Welcome to MySQL Workbench
Setup New Connection
Connection Name: big c
Type a name for the connection
Connection Method: Standard (TCP/IP)
Method to use to connect to the RDBMS
Parameters SSL Advanced
Hostname: 0kwmsgiu.us-east-1.rds.amazonaws.com Port: 3306
Name or IP address of the server host - and TCP/IP port.
Username: admin
Name of the user to connect with.
Password: Store in Vault ... Clear
The user's password. Will be requested later if it's not set.
Default Schema:
The schema to use as default schema. Leave blank to select it later.
Configure Server Management...
Test Connection
Cancel
OK


MySQL Workbench
CREATE DATABASE ex;
USE ex;
CREATE TABLE st(
table_id INT PRIMARY KEY,
stud_id INT,
stud_name CHAR(10)
);
DESC st;
Result Grid
Field Type Null Key Default Extra
table_id int NO PRI NULL
stud_id int YES NULL
stud_name char(10) YES NULL
No object selected
Action Output
1 10:19:10 CREATE DATABASE ex 1 row(s) affected 0.266 sec
2 10:19:41 USE ex 0 row(s) affected 0.203 sec
3 10:19:54 CREATE TABLE st(table_id INT PRIMARY KEY, stud_id INT, stud_name CHAR(10)) 0 row(s) affected 0.250 sec
4 10:20:05 DESC st 3 row(s) returned 0.219 sec / 0.000 sec

Charis Fernandes, 26

Query 1

Limit to 1000 rows

```sql
CREATE DATABASE ex;
USE ex;
CREATE TABLE st(
table_id INT PRIMARY KEY,
stud_id INT,
stud_name CHAR(10)
);
DESC st;
INSERT INTO st(table_id, stud_id, stud_name)
VALUES (1,100,'xyz');
SELECT * FROM st;
DELETE FROM st WHERE table_id=1;
DROP TABLE st;
```

Output

Action Output

#	Time	Action	Message
1	10:19:10	CREATE DATABASE ex	1 row(s) affected
2	10:19:41	USE ex	0 row(s) affected
3	10:19:54	CREATE TABLE st(table_id INT PRIMARY KEY, stud_id INT, stud_name CHAR(10))	0 row(s) affected
4	10:20:05	DESC st	3 row(s) returned
5	10:24:37	INSERT INTO st(table_id, stud_id, stud_name) VALUES (1,100,'xyz')	1 row(s) affected
6	10:24:46	SELECT * FROM st LIMIT 0, 1000	1 row(s) returned
7	10:25:05	DELETE FROM st WHERE table_id=1	1 row(s) affected
8	10:25:13	DROP TABLE st	0 row(s) affected

Delete namedb instance

Permanently delete **namedb** DB instance. You can't undo this action.

Proceeding with this action will delete the instance with all its content and can affect related resources. Learn more

Create final snapshot
Determines whether a final DB Snapshot is created before the DB instance is deleted.

Final snapshot name
The identifier of the new DB snapshot that is created.

namedb-snapshot

Retain automated backups
Determines whether retaining automated backups for 1 day after deletion

You will be billed for retained backup storage at the rate described as 'Additional backup storage' found in Backup Storage.

To avoid accidental deletion provide additional written consent.

To confirm deletion, type *delete me* into the field.

delete me

Cancel Delete